Journal of Engineering Research 13 (2025) 898–908

Contents lists available at ScienceDirect

Journal of Engineering Research

journal homepage: www.journals.elsevier.com/journal-of-engineering-research

Study the thermal response of a solar flat-plate collector under transient solar radiation experimentally and numerically

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ARTICLE INFO

Keywords:
Solar radiation
Computational fluid mechanics (CFD)
Surface absorption
Solar Flat Plate Collector (SFPC)

ABSTRACT

The target of the current study is to investigate and comparing the thermal response of solar flat plate collector (SFPC) for traditional and new model numerically and experimentally. The new model consists of elliptical-shaped riser pipes and a flat plate with longitudinal channels and wavy fins. The idea behind the new model is to increase the surface area exposed to solar radiation. All numerical models were created and simulated using ANSYS Fluid 19. R3. While, all of the experimental tests were conducted during the winter season in Baghdad, Iraq. The study showed that there are agree well between experimental and numerical results, and both models have the same behavior. The results showed that the new model outperformed the traditional one in terms of thermal response. The new one performed 23.6% better to compare the conventional model in terms of liquid temperature in the container. Additionally, the mass flow rate of liquid flowing within the closed loop of the new model is 7.9% higher than the traditional model. Moreover, in terms of overall thermal efficiency, the new model outperformed the traditional model by 22.4%.

1. Introduction

Nowadays, due to the rising cost of fossil fuels and the possibility of climate variation, renewable energy has gained popularity worldwide. Renewable energy is of economic and environmental benefit to humanity and can be described as clean energy that comes from natural sources. There are numerous kinds of renewable energy sources; however, a solar energy source is the most significant due to its global availability [1]. There are a variety of devices that can be used to convert solar energy into thermal or electrical energy for heating water or buildings, cooking food, etc. In order to improve the thermal response of such systems, many researchers have studied the effects of several parameters. Badiei et al. [2] carried out a numerical investigation of SFPC, with and without fins, in order to enhance their thermal response. They reported that a model with fins performed 4% better than one without fins in terms of thermal efficiency. A computational and experimental investigation was carried out by Jasim et al. [3] to examine the influence of various factors on the thermal response of the FPSC. These parameters are the addition of toroidal fins around the riser-pipe and the insertion of a closed tube within the riser-pipe, while keeping the same amount of the working liquid. They found that the thermal response of the proposed model is 25% higher than the conventional model in terms of water temperature in the container. Other researchers have tried to improve the thermal response of SFPC by inserting various tube configurations within the riser pipe. Freegah [4] studied the effect of inserting several tube configurations inside the riser pipe, namely straight, helical, and wavy, on the thermal response of the SFPC. He found that the thermal response of the straight tube model is the best among other models and is higher by 12.3% as compared to the traditional one. The researcher Jasim et al. [5] also reached the same results as Freegah, as they conducted a study on the influence of inserting a straight closed-tube into the ascending tube. They reported that the proposed model outperformed the conventional model by 15.14% and 19.89% in terms of the water temperature in the container and the working liquid temperature of the riser pipe outlet, respectively. To increase the surface area of the collector and hence increase the amount of heat that is received from the solar, Alkhafaji et al. [6] studied the effect of adding fins to the collector on the thermal response of the SFPC. They discovered that adding fins improved performance by approximately 20.8% higher than the conventional model. On the other hand, Junaid and Sadaq [7] investigated the effect of inserting fins inside the riser tube on the thermal response of the FPSC numerically. They found that adding fins to the riser pipes significantly improved the thermal

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https://doi.org/10.1016/j.jer.2024.03.004
Received 23 June 2023; Received in revised form 18 August 2023; Accepted 11 March 2024
Available online 12 March 2024
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Nomenclature

Pr	Prandtl number
SWHs	Solar water heating system
FPSC	Flat Plate Solar Collector
T	Temperature, °C
$T_{w,t}$	Temperature in tank, °C
$T_{w,o}$	The riser pipe outlet temperature, °C
q	Heat flux, w/m2
Q_u	Actual collected useful energy, w
C_p	Specific heat,J/kg.°C
ρ	Density, kg/m3
μ	Dynamic viscosity, N•s/m2
IT	Intensity of solar radiation
δ	Declaration angle
ε	Earth's orbit correction factor
θ	Incline angle of FPSC
τ	Atmospheric transmittance
ω	Sun hour angle
∝	local latitude
$\dot{m}$	Mass flow rate, kg/s
η	Thermal efficiency of the solar collector, %

Greek symbols

v	Kinematic viscosity, m2/s

efficiency and outlet temperature as compared with a conventional model, by 2 oC and 25%, respectively. Koholé & Tchuen [8] studied the effect of three types of FPSC, namely, upper, lower, and side bond configurations of absorber pipes, on the thermal response of FPSC. They found that, among all the studied models, the best one is the upper bond model of the absorption tube. Qasim et al. [9] have done numerical comparison study between two models, namely the traditional model (riser pipes are in the form of a straight) and the new model (the riser pipes are in the form of a helical). The results show that the helical riser pipe model performed best as compared to the straight pipe one. Mandal and Ghosh [10] have tried to enhance the thermal response of SFPC by studying the effect of a double riser pipe with a reflector and comparing it to a single riser pipe. They reported that, compared to a single pass, the double pass performed better. The comparison between the new model (fins inside the riser pipe) and the traditional one has been conducted experimentally by Seco-Nicolás et al. [11]. They reported that the difference in the average working liquid temperature between the two models is minimal. Gunjo et al. [12] conducted an experimental and numerical study to improve the thermal efficiency of SFPC for various operating parameters such as the flow rate of the working liquid, surrounding temperature, solar collector conductivity, solar radiation, and inlet working liquid temperature. They discovered that increasing all of the above parameters except the working fluid's inlet temperature increased the SFPC's thermal response. Shrirao et al. [13] compared the cross sections of two types of riser pipe: airfoil and circular. They found that the airfoil absorber tubes gained a large amount of solar radiation compared to the other absorber tubes. Thus, the efficiency of the airfoil-shaped absorption tube improved by 10–12% as compared to the circular cross section. Hamed [14] studied the influence of ambient temperature on the efficiency by using nine pipes welded above flat plate. They found that increasing the ambient temperature by 8 °C increases FPSC efficiency by 3%. Shelke and Patil [15] conduct a numerical study to compare the thermal response of two shapes of riser pipes in FPSC, defined as elliptical and circular pipes. They confirmed that the SFPC with the elliptical pipe has a 4.17 °C higher average surface temperature than the traditional one.

Despite the abundance of (SFPC) previous work, it has been noted that the majority of these studies were conducted under steady-state heat flux boundary condition with using active methods to flow the liquid inside the system. Keeping the above in view, the present work attempts to establish the optimal thermal response of SFPC by increasing the surface area that exposed to solar radiation (without changing the

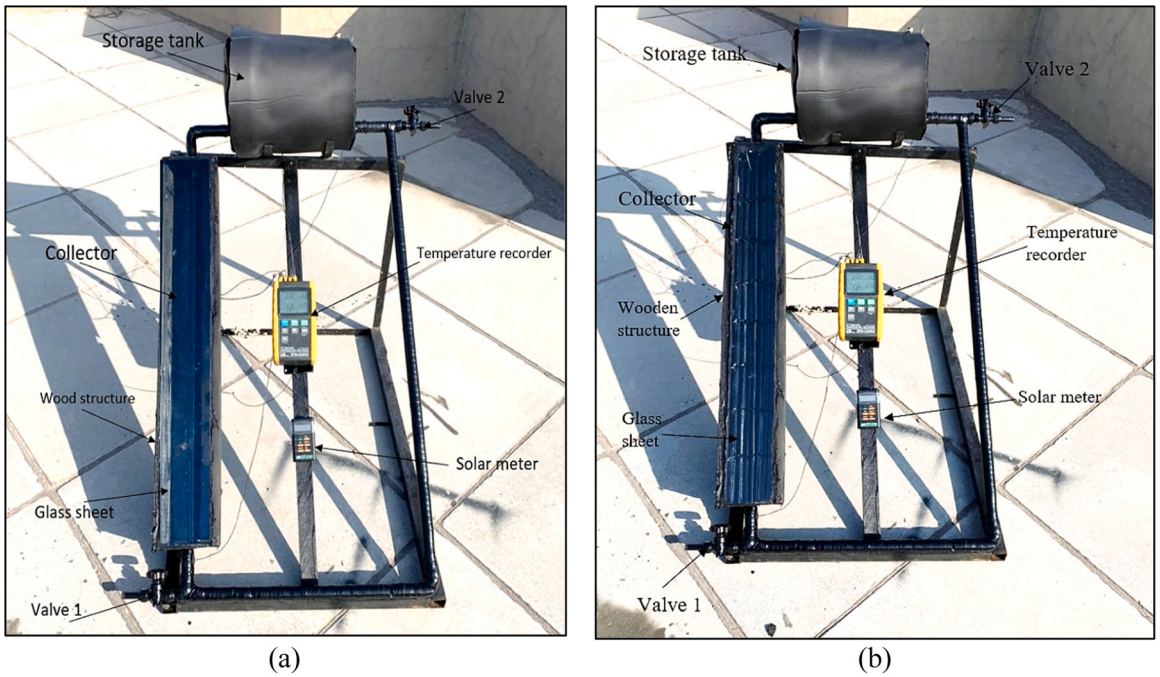


Fig. 1. Plate of the SFPC system under consideration, a) conventional model and, (b) New model.

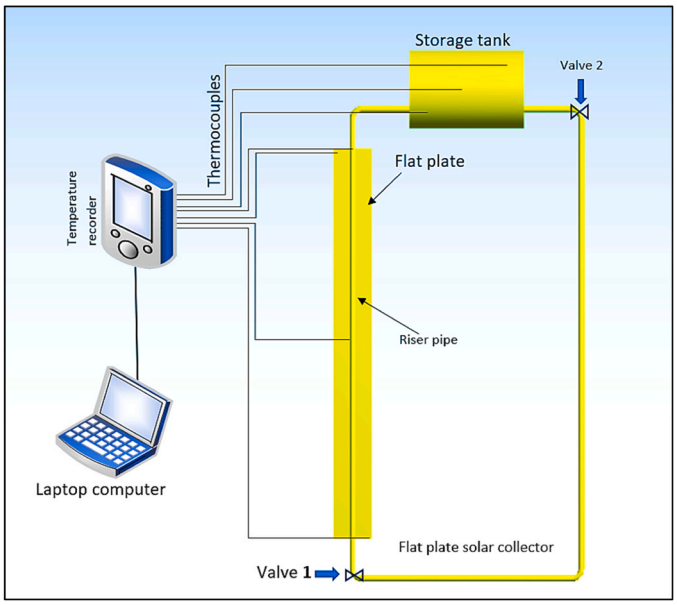


Fig. 2. Schematic of the experimental setup.

Table 1
Specifications of the experimental rig.

The parts	Specifications
The height of the SFPC	120 cm
The width of the S FPC	60 cm
tilt angle	33°
The number of riser-pipe	One
Length of the riser pipe	100 cm
Riser-pipe material	Copper
The container diameter	20 cm
The high of the container	30 cm
The material of the container	Plastic
The thickness of the rubber buffer	0.01 m
The flat plate length	1 m
Width of flat plate	0.1 m
Flat plate thickness	0.65×10^{-3} m
Data acquisition	BTM-4208 SD with 12 channels
Solar power meter	TES-1333R

collector main dimensions). Therefore, under transient heat flux and passive technique to circulate the liquid, this work aims to develop and test a new SFPC model (against traditional circular – riser pipe and smooth plate model). This new developed model contains elliptical cross-section riser pipe and flat plate with both longitudinal channels and added wavy fins.

2. Experimental setup

This section includes the description of all rig parts and the requirements for the measuring instruments utilized in the current experimental work. All of the experiments performed were carried out in Baghdad, Iraq, during the winter session.

2.1. Test rig

The experimental set-up consists of several parts namely a riser pipe,

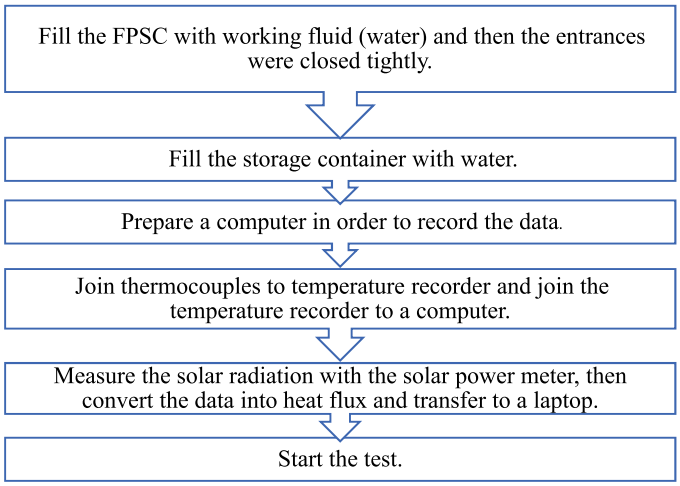


Fig. 3. Flowchart for conducting experiments.

Table 2
Standard uncertainty estimated.

	Thermocouple at the outlet of the riser (°C)	Thermocouple inside the tank (°C)	Thermocouple at the flat plate (°C)
Average standard uncertainty	$\mp$ 0.607	$\mp$ 0.225	$\mp$ 0.49

a 9.6-liter water storage container, a heat exchanger inside the container, a wooden frame, a glass layer, flat-plate, and a down-comer pipe that return the liquid to the riser-pipe as shown in Fig. 1. Further a number of temperature sensors (type K) are utilized. One is utilized to evaluate the temperature of the working fluid at the riser pipe's outlet, three are used to measure the temperature of the water inside a container, and three are employed to gauge the plate temperature. Fig. 2 depicts the operation of a laptop-connected solar meter and data acquisition device. The wooden structure is 1 m long, 0.1 m wide, and 0.1 mm thick, with a total length of 0.01 m. A copper flat plate with dimensions of 0.109 m wide by 1 m long by 0.65×10^{-3} m thick and a glass sheet with measurements of 1.006 m long by 0.130 m wide by 6×10^{-3} m thick are used to lid the collector, with an air gap of 0.3 m above the absorber plate, respectively. In order to prevent the loss of heat through the air gap and fix the glass in place, rubber was used. Furthermore, all experimental test data was saved on a laptop. 2 In addition, the descriptions of the test rig used in the present study are depicted in Table 1.

2.2. Procedure of test

The present experimental test was done on the 15th of November outdoors for period between 8:00 am. and 3:00 pm. Before beginning the experimental test, several steps should be taken into account. These steps are summarized in a flow chart as shown in Fig. 3.

2.3. Estimating uncertainty

Estimating uncertainty is considered as the amount of fluctuations or error measurements because of user error. Such errors can adversely affect both the working fluid flow and the temperature measurement of the FPSC. Measurement errors are calculated by many tools, which cannot work under the ideal measuring conditions.

To calculate uncertainty, the first step is to calculate the mean sample. The mean sample $\overline{X}$ is calculated by taking the sample average

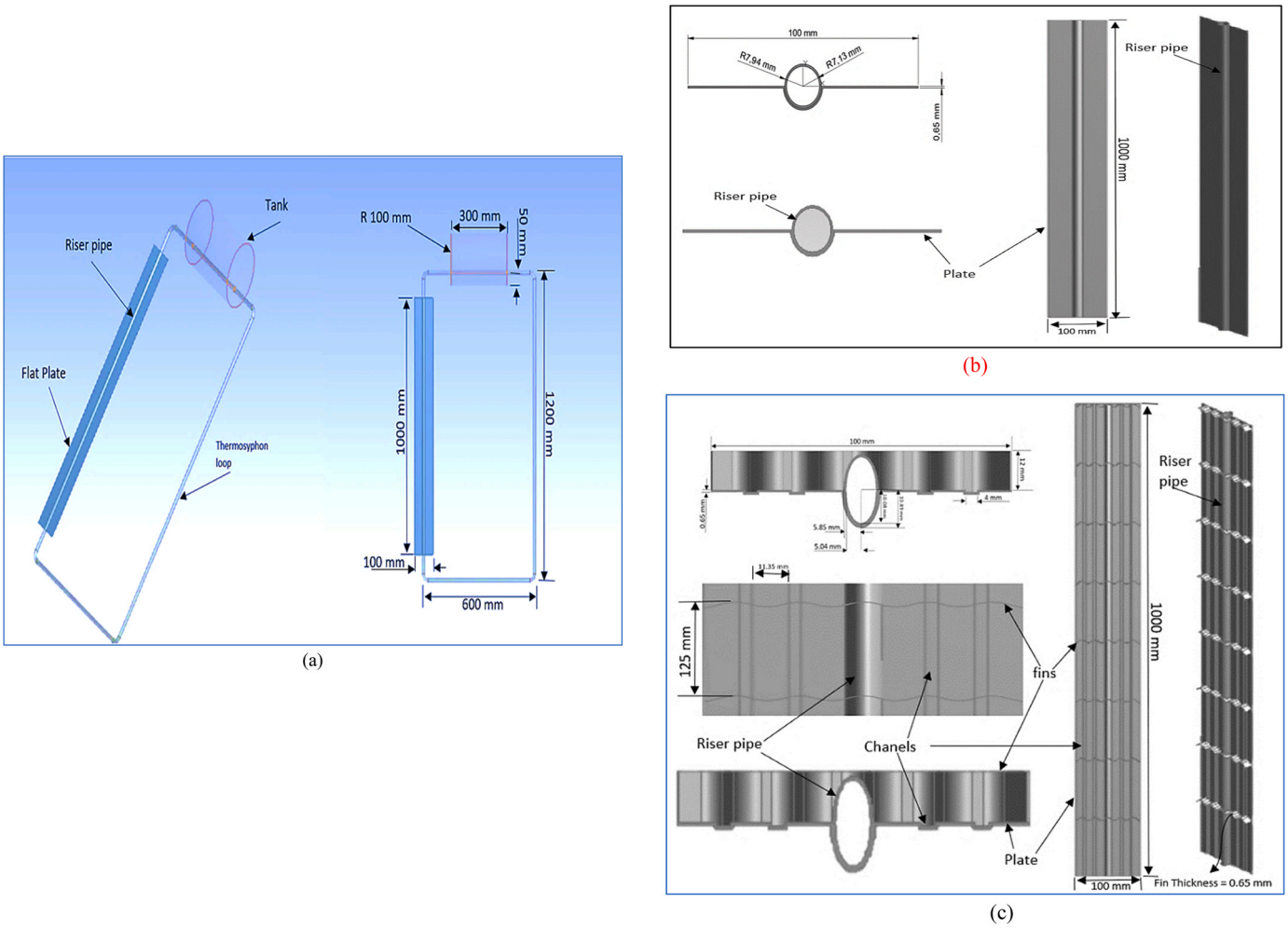


Fig. 4. Geometry of SFPC for (a) the SFPC whole system, (b) the conventional collector, and (c) the new collector.

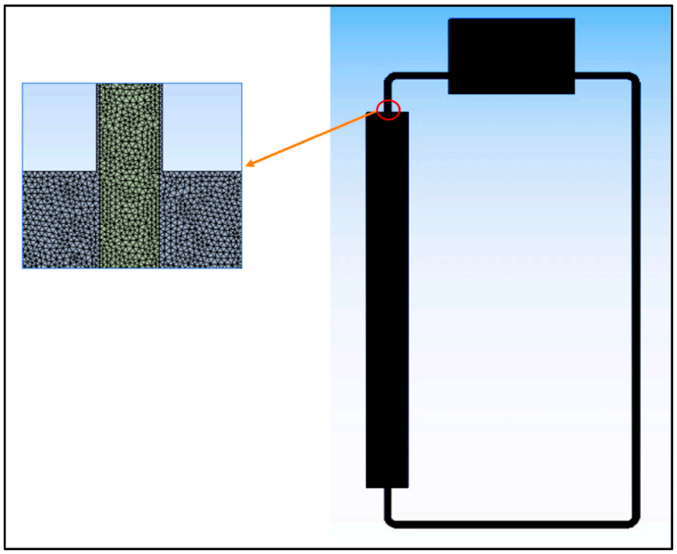


Fig. 5. SFPC meshing.

Table 3
Mesh independency.

Model	Element no. (million)	Water of container temperature (°C)	Percentage Difference (%)
Conventional	2.21	26.67	0.22
	4.51	26.61	
New model	3.4	34.85	0.229
	6.9	34.93	

value. The average value is calculated using Eq. (1) by combining the sampled values and dividing them by the sample size n [16].

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{1}$$

Where,$\overline{X}$, X_i and n are respectively mean value, sample value and sample number.

Then, the sample standard deviation was computed. The sample standard deviation (S_X) was computed using Eq. (2) via taking the sum square root of the sampled deviations squares from the mean divided via (sample number −1) [17].

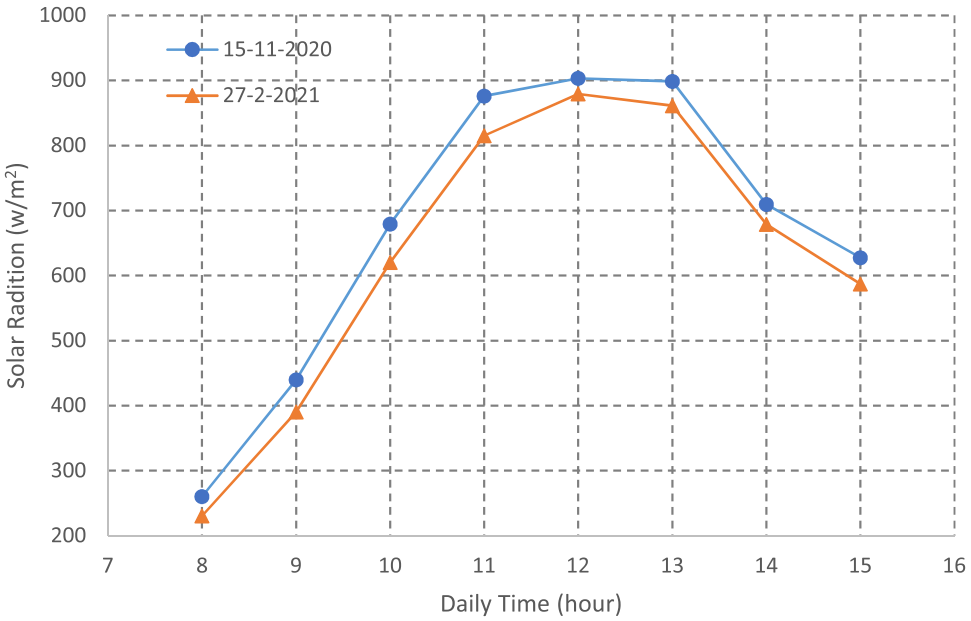


Fig. 6. Variation in solar radiation for seven hours.

$$S_X = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2} \quad (2)$$

Where, n-1 is the degrees of freedom for the estimate. The degrees of freedom used to estimate the uncertainty are important for determining the confidence limits and the other decision variables. Lastly, Eq. (3) can be used to estimate the standard uncertainty. Based on the Eqs. (1), (2) and (3), the uncertainty is calculated and the results are listed in Table 2.

$$Eu = \frac{S_X}{\sqrt{n}} \quad (3)$$

3. Numerical analysis

In order to analyze any model with ANSYS Fluent, several steps must be taken. The first step involves creating the model and generating the mesh domain. The second step includes determining the boundary conditions and setting them up. At the third step, the outcomes are presented. In this study, ANSYS Fluent R 19.3 is used.

3.1. Geometric models

In this study, two models of the SFPC are created, namely the traditional model (circular cross-sectional shape of the riser pipe, smooth plate, and without fins) and the new model (elliptic cross-sectional shape of the riser-pipe, the plate with a channel, and the wavy fins on the flat plate), as shown in Fig. 4(a-c). A SFPC collector has a width and height of 60 cm and 120 cm, respectively, with a incline angle of 33°, with one riser pipe connecting the bottom with the up-riser. Furthermore, the riser pipe's length for both models is 1 m, while the thickness and diameter of the return pipe have similar dimensions to the riser-pipe of the traditional one.

3.2. Mesh generation

A mesh was created for the system, where the hybrid meshing has been used to simulate the model after completing the engineering model process, as shown in the Fig. (5). The hybrid mesh has two hexagonal forms and tetrahedral each type has parameters, characteristics, and conditions to be used. Compared to other methods, the hexagonal mesh is considered one of the most widespread methods because it has many advantages, such as less memory needs, provides further accurate outcomes because of the less diffusion, and the capability to connect the complex engineering shapes [18].

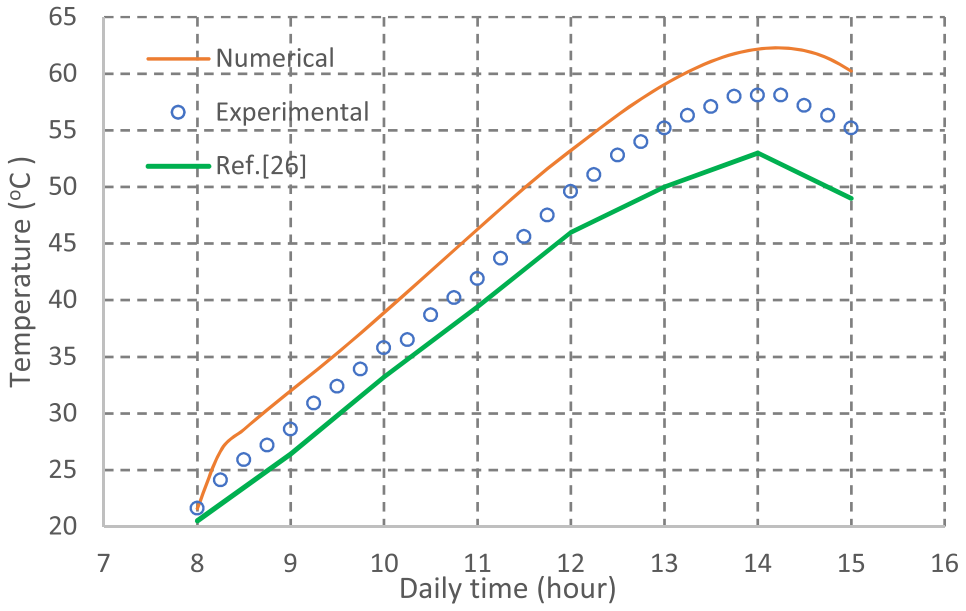
Furthermore, the element number of the grid has an important influence on the accuracy of the numerical results. So mesh independency tests were performed in order to select the sufficient element number of the mesh. Table 3 shows the findings of these tests, which summarize the numerical outcomes of the average water temperature in the container at various mesh shapes for both models under study. Based on Table 3, the doubled of the element numbers had little effect on the numerical results. So, the minimum number of elements for all models in this study was used to reduce the cost and the simulation run time.

3.3. Boundary conditions

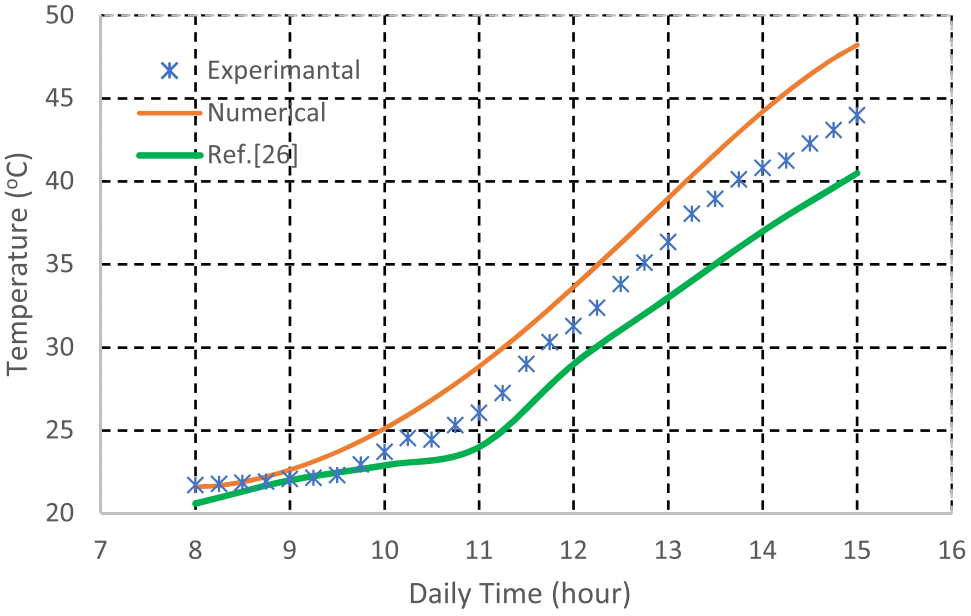
One of the boundary conditions used in the current study is the initial temperature of water in the container and the working liquid temperature inside the pipe loop of 21.6 °C. Furthermore, the container contains 9.6 L of water, and the closed loop contains 0.6 L of working fluid. The riser-pipe and plate were exposed to transient heat flux for seven hours (see Fig. 6), which calculated from Equ. 4 [19]. To input the heat flux values to the Ansys Fluent software, a user-defined function (UDF) was used ANSYS (FLUENT) as the input value for heat flux (CFD). The water that circulated inside closed loop was water. The physical properties of water used in the numerical aspect are density (998.2 kg/m^3), heat capacity (4.182 kJ/kg-K), viscosity (1003 *10^{-6} kg/m.s), thermal conductivity (0.6 W/m.K) and thermal expansion rate (149 *10^{-6}1/K).

$$q = I_T \varepsilon \tau [\ \sin\delta\sin\ (\theta - \alpha) + \cos\delta\cos\ (\theta - \alpha)\cos\omega\ ] \quad (4)$$

where δ,τ,θ, I_T,α and ω denote to the tilt angle of SFPC, the atmospheric transmittance, local latitude, solar radiation intensity and the hour angle of the sun respectively. By using Eq. (5), it can be estimate the



(a)



(b)

Fig. 7. Validation of numerical outcomes for (a) working liquid temperature at the riser-pipe exit and (b) the fluid temperature inside container storage.

Table 4
Acceptance criteria of statistical tests.

Test type	Criteria of acceptance	The result of validation
Chi-square	Less than 0.05 is acceptable [26]	0.012
t-Test	Closed to 0 [27]	0.000015
Durbin–Watson statistic	Less than two is acceptable [28]	1.2104

correction factor for Earth's orbit ε [20].

$$\varepsilon = \left[1 + 0.033\cos\left(\frac{360N_d}{365}\right)\right] \quad (5)$$

N_d depicted the number of the day over the year.

3.4. Governing equations

The 3D Navier-Stokes equations, the energy equation, and the continuity equations were numerically resolved.

The continuity equation governing incompressible fluid flow can be written as below:

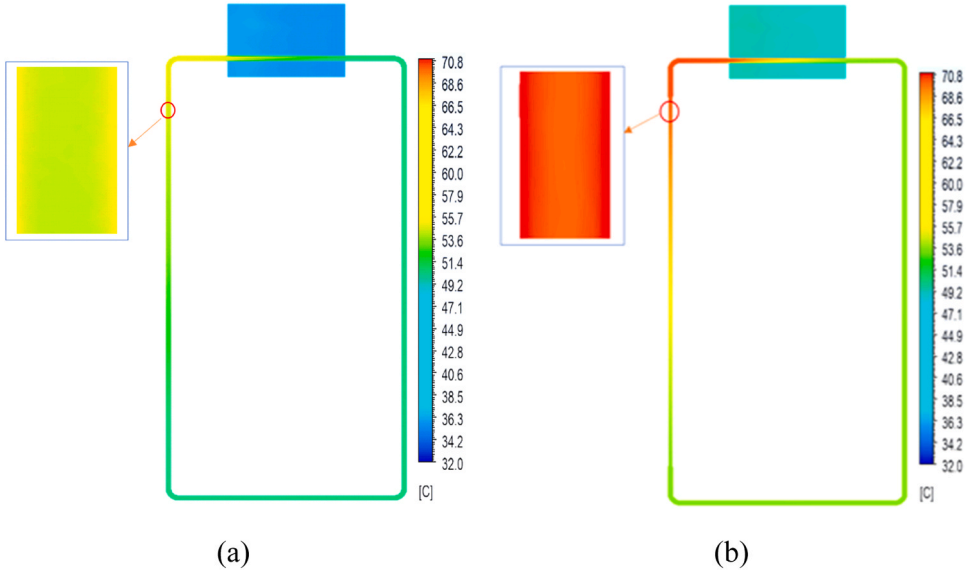


Fig. 8. Temperature contour of the water in the container and the working liquid for both models: (a) the traditional one and (b) the new one.

$$\frac{\partial}{\partial x}\quad(\rho u_i) = 0 \tag{6}$$

The following equation can be used to calculate heat conduction in 3D solid materials: [21]

$$K_m\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)=0 \tag{7}$$

Heat convection can be estimated using Eq. 8 [22].

$$\nabla.\left(\rho h\overrightarrow{U}\quad\right) = -p\nabla\overrightarrow{U}+\nabla(k\nabla T)+\phi+s_h \tag{8}$$

The 3-D of the Navier-Stokes equations can be written as in Eq. (9) [23]:

$$\begin{aligned}
\nabla.\left(\rho\overrightarrow{U}\quad u\right) &= -\frac{\partial p}{\partial x}+\frac{\partial \tau_{xx}}{\partial x}+\frac{\partial \tau_{yx}}{\partial y}+\frac{\partial \tau_{zx}}{\partial z}\\
\nabla.\left(\rho\overrightarrow{U}\quad v\right) &= -\frac{\partial p}{\partial y}+\frac{\partial \tau_{xy}}{\partial x}+\frac{\partial \tau_{yy}}{\partial y}+\frac{\partial \tau_{zy}}{\partial z}\\
\nabla.\left(\rho\overrightarrow{U}\quad w\right) &= -\frac{\partial p}{\partial z}+\frac{\partial \tau_{xz}}{\partial x}+\frac{\partial \tau_{yz}}{\partial y}+\frac{\partial \tau_{zz}}{\partial z}
\end{aligned} \tag{9}$$

3.5. Computation of the collector thermal efficiency

The instantaneous thermal efficiency of a solar collector is defined as the ratio of the collected useful energy (Qu) to the intensity of the solar radiation. It is calculated as follows [24]:

$$\eta = \frac{Q_u}{A_c I_T} \tag{10}$$

Where, the instantaneous useable energy gathered via the solar collector is specified via:

$$Q_u = \dot{m}C_p(T_{w,o}-T_{w,i}) \tag{11}$$

Where, ṁ, C_p, $T_{w,o}$, and $T_{w,i}$ represent the mass flow rate of working liquid, the specific heat of water, the outlet and the inlet of water temperature, respectively.

4. Result and discussion

The normal convection phenomenon was demonstrated using an Ansys Fluent numerical analysis. Furthermore, the average temperature of the fluid in the container and the SFPC loop were used to calculate the fluid properties.

4.1. Numerical approach validation

This section shows a comparison between experimental and numerical outcomes. This comparison was conducted to prove the validity of the numerical model. This comparison was done for the temperature of the container water and the working liquid at the riser-pipe outlet, as shown in Fig. 7. It is clear from this Fig. that the maximum difference between the numerical and experimental results for each of the average temperatures of the container water and the working liquid are 8.12% and 8.36%, respectively. From another side the current results have been compared with previous experimental study Ref. [25]. This compared shown that the results of the current study have same behavior of previous experimental resulst. Furthermore, some statistical analysis to prove that the results are very close has been done, and the results of these statistical analyses are shown in Table 4. These estimates give a good indication that the developed numerical approach is reliable for investigating the heat transfer improvements of the new model.

4.2. Numerical analysis

Fig. 8 shows the temperature contour at 13 p.m. for the conventional and new models for working liquid within the pipes loop and water within the container. This Fig. clearly shows that the working liquid temperature inside the riser-pipe part is higher than the down-comer part for both models. Furthermore, the new model's average water temperature inside the container and working liquid temperature are higher than the conventional model's. This is due to the increased absorption area that the collector of new model has as a result of changing the cross-sectional shape of the riser pipe to elliptic, adding fins to the plate, and making a channel on the plate. In addition, the average temperature of the container water is 38.99 °C for the conventional

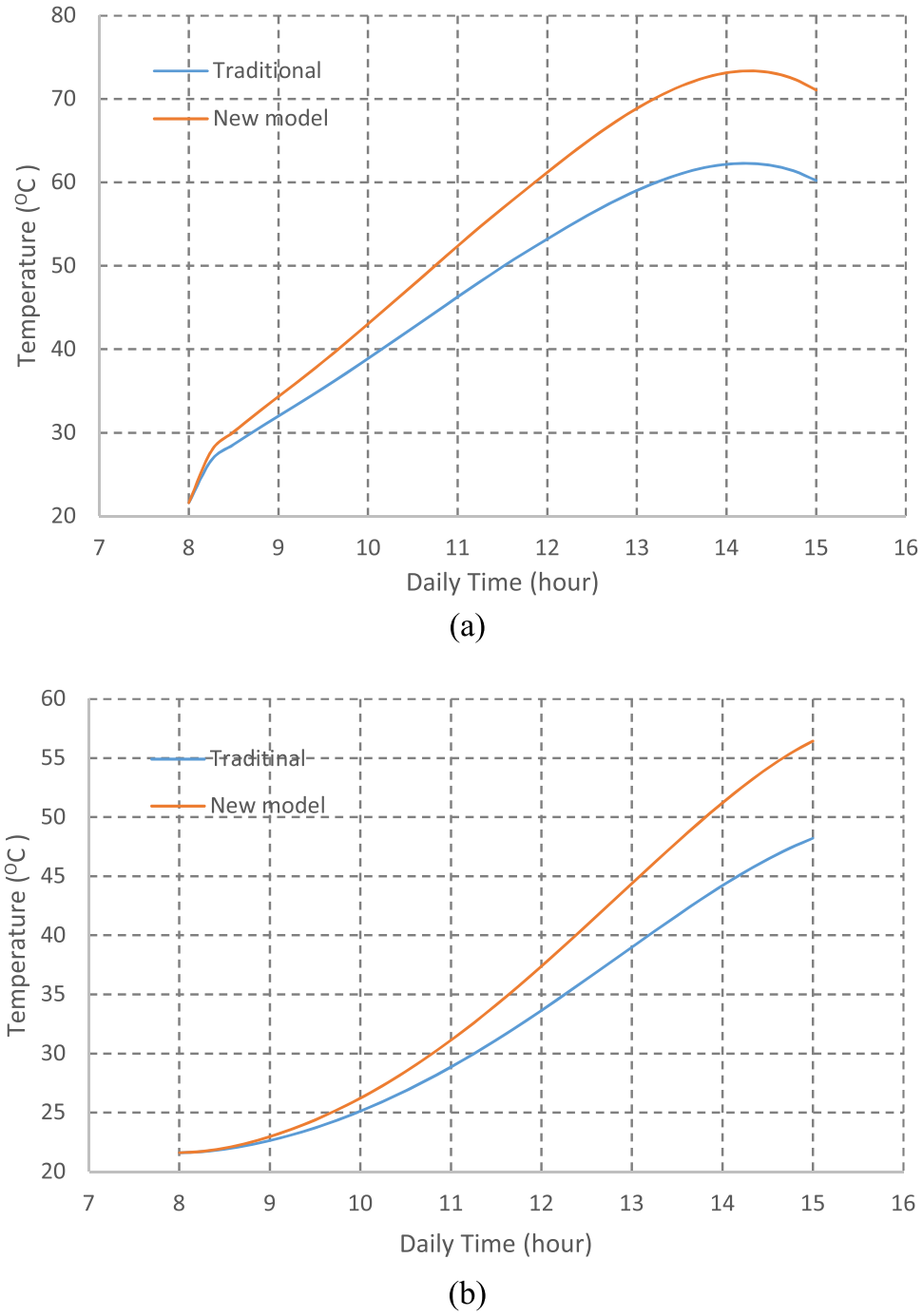


Fig. 9. Temperature distribution of both models (a) the working liquid at the upriser and (b) the container water.

model and 44.39 °C for the new model, respectively.

Fig. 9 depicts the temperature variation for (a) the working liquid at the upriser and (b) the container water for both models considered in this study. Fig. (9, a) depicts that the maximum working liquid temperature at the upriser-pipe is at 14 p.m., while the temperature decreases in the last hours as heat flux decreases. The temperature of the water at the upriser has increased by 40.69 °C and 51.78 °C, for new and conventional models, respectively. Fig. (9, b) shows that the water temperature in the container increased by 26.62 °C for the conventional model and 34.85 °C for the new model, respectively. It can be concluded that the new model performed better than the traditional one in terms of the temperature of the container water and the working liquid temperature at the riser-pipe outlet. This is because an increase in the collector surface area of the new model.

Fig. 10. depicts the variation of the working liquid mass flow rate for both models. The working liquid flow rate inside the riser pipe of the SFPC increases and then decreases as a result of the increase and decrease in the rate of solar radiation. The maximum value of the flow rate for both models happens at one o'clock in the afternoon. The flow rates for the conventional, and new models are 1.29 g/s and 1.403 g/s respectively.

Fig. 11 illustrates the variation in thermal efficiency of SFPCs during operating time for both models. This Fig. shows that as solar radiation increases and decreases, the thermal efficiency of the SFPC increases, reaching a maximum at midday and then decreasing at the end of the day. At 13 pm, SFPC the thermal response is 59.8% and 77.05% for

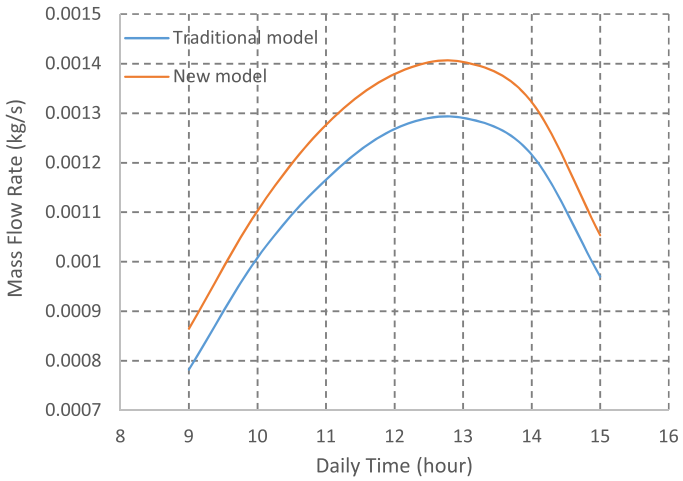


Fig. 10. The variation of the working liquid mass flow rate distribution at the upriser pipe outlet.

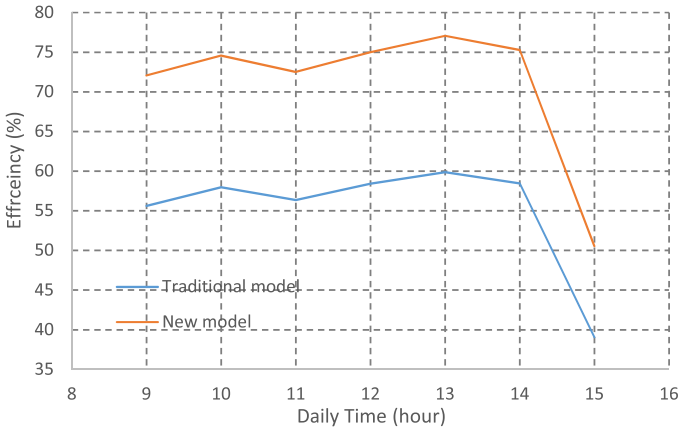


Fig. 11. Instantaneous thermal efficiency variance for SFPC.

traditional model and new model respectively. Meanwhile, Fig. 12 depicts the overall efficiency of SFPC for conventional, and new models. According to this Fig., the new model has the highest overall thermal efficiency as compared to the conventional model, and this could be related to the increased absorption surface area of the SFPC. As the overall thermal efficiency value of the traditional and new models is 55.1% and 71%, respectively.

4.3. Numerical validation of the new model

The same boundary conditions and geometrical dimensions are used in the experimental test rig to validate the numerical results. Upon such base, the temperature of water in the container and the temperature of working liquid at the riser pipe outlet for (7 hours) operating are compared as displayed in Fig. (13) (a & b). This Fig. shows that the numerical temperature behavior is the same as the experimental behavior. In addition, close results are found between the experimental and the numerical findings. For the average temperature of the container water and the working liquid at the riser-pipe outlet, the difference between the numerical and experimental results is 9.42% and 10.46%, respectively, as shown in Fig. 13 (a and b). There're many causes for the difference between the experimental outcomes and the present results. Nevertheless, the thermal losses happened during the experimental test that being hard for monitoring, and the errors resulted via the inadequate precision of measuring apparatuses are among these most significant causes.

Finally, it can be summarized that the comparison between this study and previous studies similar to the present study in terms of the overall efficiency of SFPC enhancements as shown in Table 5.

5. Conclusions

In the current work, the thermal response of SFPC for the two models under consideration was investigated numerically and experimentally. Under the limitations of the current study, the following conclusions can be drawn:

1. The disparity between the numerical and experimental findings was found to be 8.12% and 8.36%, for the average temperature of the water in the container and the working liquid at the outlet of the riser pipe for the conventional model. While for the new model, the disparity between the numerical and experimental findings was found to be 9.42% and 10.46%, respectively, for the working liquid at the outlet of the riser pipe and the average temperature of the container water.
2. Compared to the conventional model, the new model performed 23.6% better for water temperature in the container and 21.9% better for working liquid temperature at the riser pipe outlet.
3. In the new model, the working liquid mass flow rate is 7.9% higher than in the conventional model.

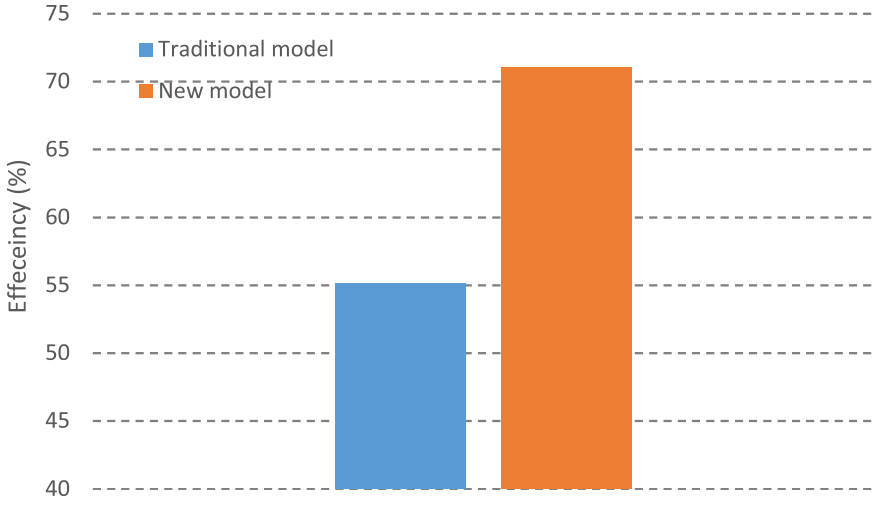


Fig. 12. The overall thermal efficiency for SFPC.

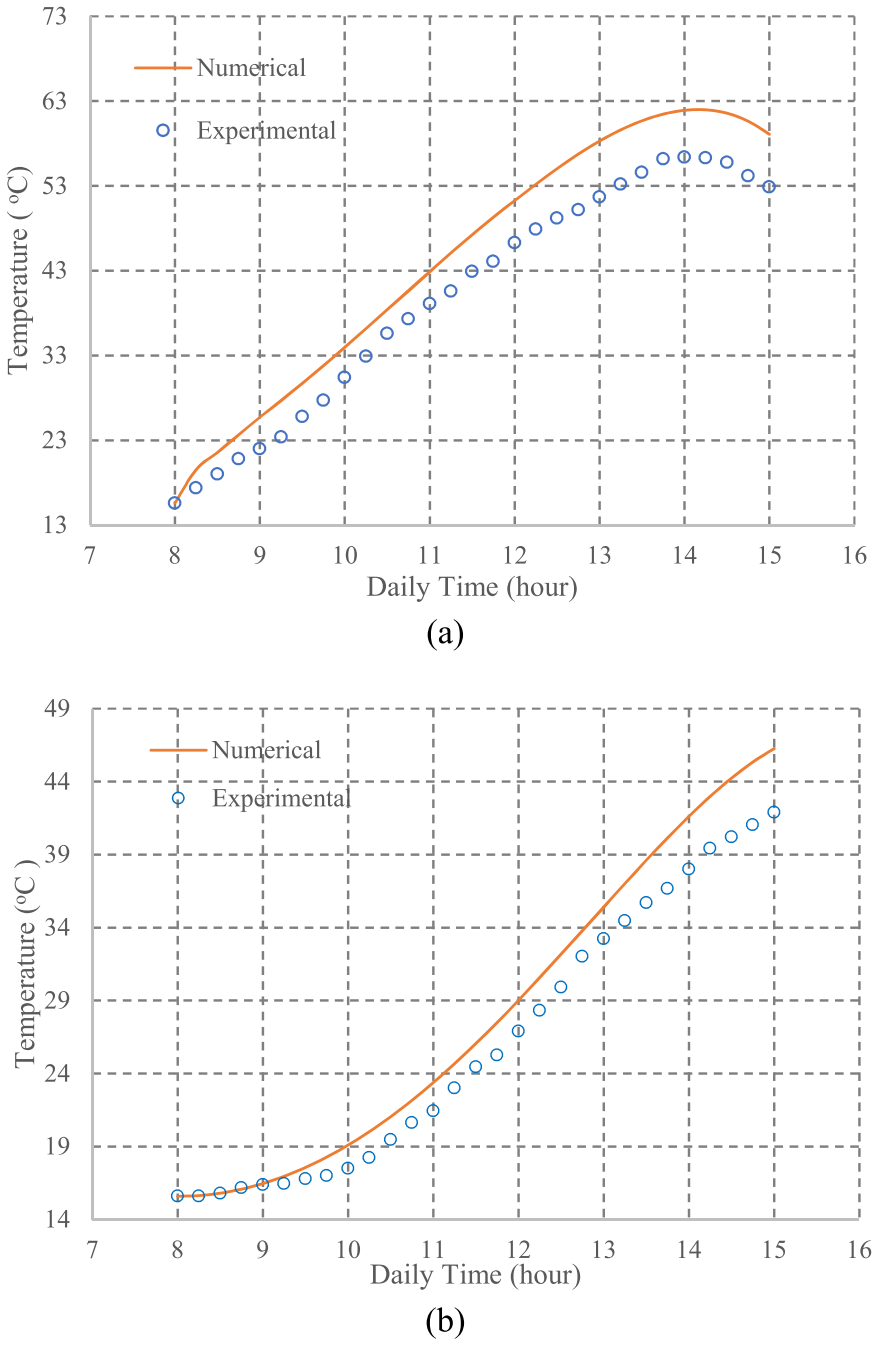


Fig. 13. Comparison between the numerical and experimental findings of the temperature (a) the working liquid at the riser-pipe outlet and (b) the container water.

Table 5
The percentage of the overall efficiency of SFPC enhancements of the current study with previous studies.

	The best model of the present study	Herrero et al. [29]	Vijayakumar et al. [30]
The overall efficiency of SFPC	22.4%	4.5%	18.6%

4. The new model performed 22.4% better than the conventional model in terms of the overall efficiency of SFPC.

In order to establish the optimal thermal response of SFPC, this study provides the necessary knowledge base.

Declaration of Competing Interest

Also the authors declare no conflict of interest.

Acknowledgment

The authors would like to thank Mustansiriyah University (www.

uomustansiriyah.edu.iq) in Baghdad -Iraq for its support in the present work.

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